Solution

Matrix Theory, Quiz 9W April 24, 2026

Perform the Gram-Schmidt process on the set

$$\mathbf{x}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix}2\\-3\\-2\end{bmatrix},$$

to construct an orthogonal basis for $\mathbb{R}^3$.

2 Step 1. $V_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}$ $A_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}$ $D_1 = A_1^T A_1 = (3)$

3 Step 2. $V_2 = X_2 - A_1 D_1^{-1} A_1^T X_2 = X_2 - \begin{bmatrix}1\\1\\1\end{bmatrix}(1/3)(1,1,1)\begin{pmatrix}1\\2\\3\end{pmatrix} = X_2 - \begin{bmatrix}1\\1\\1\end{bmatrix}(6/3)$

$= \begin{bmatrix}1\\2\\3\end{bmatrix} - \begin{pmatrix}2\\2\\2\end{pmatrix} = \begin{pmatrix}-1\\0\\1\end{pmatrix}$

$A_2 = \begin{bmatrix}1 & -1\\1 & 0\\1 & 1\end{bmatrix}$ $D_2 = A_2^T A_2 = \begin{bmatrix}3 & 0\\0 & 2\end{bmatrix}$

4 Step 3. $V_3 = X_3 - A_2 D_2^{-1} A_2^T X_3 = X_3 - A_2 D_2^{-1}\begin{bmatrix}1 & 1 & 1\\-1 & 0 & 1\end{bmatrix}\begin{bmatrix}2\\-3\\-2\end{bmatrix}$

$= X_3 - A_2\begin{bmatrix}1/3 & 0\\0 & 1/2\end{bmatrix}\begin{pmatrix}-3\\-4\end{pmatrix} = X_3 - \begin{bmatrix}1 & -1\\1 & 0\\1 & 1\end{bmatrix}\begin{bmatrix}-1\\-2\end{bmatrix}$

$= \begin{bmatrix}2\\-3\\-2\end{bmatrix} - \begin{bmatrix}1\\-1\\-3\end{bmatrix}$

$= \begin{pmatrix}1\\-2\\1\end{pmatrix}$

1 $A = \begin{bmatrix}1 & -1 & 1\\1 & 0 & -2\\1 & 1 & 1\end{bmatrix} \leftarrow$ orthogonal matrix